

IDENTITÉS FONDAMENTALES

$$\begin{aligned}\cos^2(x) + \sin^2(x) &= 1 \\ 1 + \tan^2(x) &= \frac{1}{\cos^2(x)}\end{aligned}$$

FORMULES D'ADDITION

$$\begin{aligned}\cos(a+b) &= \cos(a)\cos(b) - \sin(a)\sin(b) \\ \cos(a-b) &= \cos(a)\cos(b) + \sin(a)\sin(b) \\ \sin(a+b) &= \sin(a)\cos(b) + \cos(a)\sin(b) \\ \sin(a-b) &= \sin(a)\cos(b) - \cos(a)\sin(b) \\ \tan(a+b) &= \frac{\tan(a)+\tan(b)}{1-\tan(a)\tan(b)} \\ \tan(a-b) &= \frac{\tan(a)-\tan(b)}{1+\tan(a)\tan(b)}\end{aligned}$$

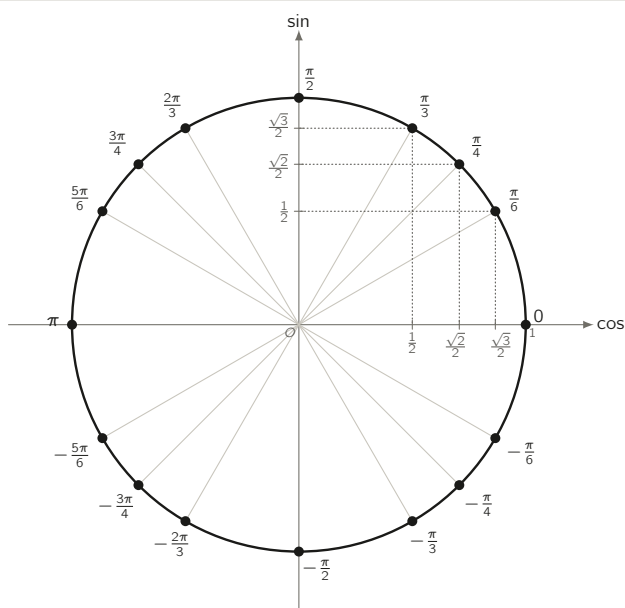
FORMULES DE DUPLICATION

$$\begin{aligned}\sin(2x) &= 2 \sin(x) \cos(x) \\ \cos(2x) &= \cos^2(x) - \sin^2(x) \\ &= 2 \cos^2(x) - 1 = 1 - 2 \sin^2(x) \\ \tan(2x) &= \frac{2 \tan(x)}{1 - \tan^2(x)}\end{aligned}$$

FORMULES D'ANGLE MOITIÉ

$$\begin{aligned}\sin^2\left(\frac{x}{2}\right) &= \frac{1-\cos(x)}{2} \\ \cos^2\left(\frac{x}{2}\right) &= \frac{1+\cos(x)}{2} \\ \tan\left(\frac{x}{2}\right) &= \frac{\sin(x)}{1+\cos(x)} = \frac{1-\cos(x)}{\sin(x)}\end{aligned}$$

CERCLE TRIGONOMÉTRIQUE



PRODUIT EN SOMME

$$\begin{aligned}\cos(a) \cos(b) &= \frac{1}{2}(\cos(a-b) + \cos(a+b)) \\ \sin(a) \sin(b) &= \frac{1}{2}(\cos(a-b) - \cos(a+b)) \\ \sin(a) \cos(b) &= \frac{1}{2}(\sin(a+b) + \sin(a-b))\end{aligned}$$

SOMME EN PRODUIT

$$\begin{aligned}\cos(p) + \cos(q) &= 2 \cos\left(\frac{p+q}{2}\right) \cos\left(\frac{p-q}{2}\right) \\ \cos(p) - \cos(q) &= -2 \sin\left(\frac{p+q}{2}\right) \sin\left(\frac{p-q}{2}\right) \\ \sin(p) + \sin(q) &= 2 \sin\left(\frac{p+q}{2}\right) \cos\left(\frac{p-q}{2}\right) \\ \sin(p) - \sin(q) &= 2 \cos\left(\frac{p+q}{2}\right) \sin\left(\frac{p-q}{2}\right)\end{aligned}$$

ANGLES ASSOCIÉS

$$\begin{aligned}\cos(\pi - x) &= -\cos(x) \\ \sin(\pi - x) &= \sin(x) \\ \tan(\pi - x) &= -\tan(x) \\ \cos(\pi + x) &= -\cos(x) \\ \sin(\pi + x) &= -\sin(x) \\ \tan(\pi + x) &= \tan(x) \\ \cos\left(\frac{\pi}{2} - x\right) &= \sin(x) \\ \sin\left(\frac{\pi}{2} - x\right) &= \cos(x) \\ \cos\left(\frac{\pi}{2} + x\right) &= -\sin(x) \\ \sin\left(\frac{\pi}{2} + x\right) &= \cos(x)\end{aligned}$$

VALEURS REMARQUABLES

x	0	$\frac{\pi}{6}$	$\frac{\pi}{4}$	$\frac{\pi}{3}$	$\frac{\pi}{2}$
$\cos(x)$	1	$\frac{\sqrt{3}}{2}$	$\frac{\sqrt{2}}{2}$	$\frac{1}{2}$	0
$\sin(x)$	0	$\frac{1}{2}$	$\frac{\sqrt{2}}{2}$	$\frac{\sqrt{3}}{2}$	1
$\tan(x)$	0	$\frac{\sqrt{3}}{3}$	1	$\sqrt{3}$	$\times$

ÉQUATIONS TRIGONOMÉTRIQUES

$$\begin{aligned}\cos(x) = \cos(a) &\iff x \equiv a [2\pi] \text{ ou } x \equiv -a [2\pi] \\ \sin(x) = \sin(a) &\iff x \equiv a [2\pi] \text{ ou } x \equiv \pi - a [2\pi] \\ \tan(x) = \tan(a) &\iff x \equiv a [\pi]\end{aligned}$$

DOMAINES, PÉRIODES, PARITÉ

cos, sin	définies sur $\mathbb{R}$	2π -périodiques
tan	définie sur $\mathbb{R} \setminus \{\frac{\pi}{2} + k\pi\}$	π -périodique
parité	cos paire	sin, tan impaires